

Luo Yan

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Supervisor: Ding Han (Academician of the Chinese Academy of Sciences),

 **Xiong Zhenhua** (Professor at Shanghai Jiao Tong University).



EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China

Sep.2021-present

Major in Mechanical Engineering, going from a Bachelor's to a Ph.D.

Relevant Coursework: Computational vision (A), Digital signal processing(A), Robot performance simulation and control principle(A-), Basis of software technique(A), Modern control theory(A-), Intelligent control technique(A-). **(GPA:3.51/4.0)**

Yanshan University (YSU), Hebei, China

Sep.2017-June.2021

Major in Automation, Bachelor. (Ranked 1st out of 373 students).

Relevant Coursework: Advanced Mathematics(B+), Linear Algebra(A), Feedback Control Theory(A), Microcomputer Principle(A+), Simulation Electronic Technology(A), Physical Fitness Test(A), Embedded single chip microcomputer principle(A+), Numerical Analysis(A).



EXPERIENCE AND SKILLS

- Responsible for intelligent manipulation and grasping of semi-humanoid robotic embodiment, including **language-vision-motion modeling**. Using **teleoperation** to acquire motion data combined with **reinforcement learning** for training.
- Responsible for Jiangsu Provincial Science and Technology Department's Frontier Leading Technology Basic Research Program and Jiangsu Province Industrial And Information Industry Transformation And Upgrading Project : **Humanoid robot** dynamic balance control algorithm research and development.
- Participated in the project of Science and Technology Development Program of Jilin Province: Research and development of robot operation system based on machine vision technology, responsible for **3D vision** disordered grasping and **vision servo**.
- Participated in China Internet+ Innovation and Entrepreneurship Competition, responsible for **quadruped robot** control algorithm development as well as China University Student Mathematical Modeling Competition, responsible for **simulation and algorithm optimization**.
- Proficient in Python, C++, C, Matlab, Blender, SolidWorks, etc. Familiar with Ubuntu, ROS, Gazebo, MuJoCo, Isaac Gym/Sim/Lab, etc., with strong self-driven learning ability.



ACADEMIC ACHIEVEMENTS

- **Luo Y**, Xiong Z, Ding H, et al. CoPickVLM: A Vision-Language Model Guided Dual-Arm Collaborative System for OcclusionAware Tomato Harvesting. *Jul.2025*
- **Luo Y**,Xiong Z,Ding H,et al. MotionVL:Visual-Language Supervision for Guiding Reinforcement Learning in Humanoid Motion[J].IEEE Robotics and Automation Letters.*Jul.2025*
- **Luo Y**, Liu C, Xiong Z, et al. DCM-based dynamic stable walking under terrain-induced

- time-varying disturbances for humanoid robots[J]. Science China Tech Sciences. *Jan. 2025*
- Pang F, Chen Y, **Luo Y**, et al. A Fast Obstacle Detection Algorithm Based on 3D LiDAR and Multiple Depth Cameras for Unmanned Ground Vehicles[J]. Drones. *Nov.2024*
 - **Luo Y**. Technical Requirements for Humanoid Robotics (**T/CIET 648–2024**). China International Economic and Technical Cooperation Promotion Association Standard. *Sep. 2024*
 - **Luo Y**, Chen G, Liu C, et al. Image Foreground Segmentation Based on Small Data Set for Visual Servo Applications[C]//2023 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM). IEEE, 2023: 715-720. *Jun.2023*
 - Chen G, **Luo Y**, Huang B, et al. BP neural network-fuzzy control-based uncalibrated visual servoing technology for robots[J]. Science, Technology and Engineering, 2023. *Mar.2023*
 - Zhang M, **Luo Y**, Chen G, et al. Design and research of intelligent maintenance robot for steam generator in nuclear power plant[J]. Science, Technology and Engineering, 2023, 23(22): 9559-9566. *Feb,2023*
 - [Invention Authorization] A Method for Remote Operation Control of a Semi-Humanoid Robot Based on VR Devices (ranking second) *Nov.2025*
 - [Invention Authorization] A visual servo method based on the mixing of multiple image feature information (ranking second) *Oct.2022*
 - [Invention Announcement] A robot automatic loading and unloading system (ranking first) *Jun.2020*
 - [Invention Announcement] An intelligent recognition handwritten mathematical formula review system (ranking first) *Nov.2019*



HONORS AND AWARDS

- Outstanding Graduate, Shanghai Jiao Tong University(SJTU) *Dec.2025*
- Selected for the Jiangsu Province Young Talent Support Program *Jul.2025*
- Senior Consultant Expert of Nanjing Production and Economy Consultant Pool *Oct.2024*
- Chief liaison officer of world famous universities of Jiangsu Innovation and Entrepreneurship Alliance *Apr.2024*
- Academic first-class scholarship of SJTU *Dec.2025&2024&2023&2022*
- Yanshan University Person of the Year (Only 5 person in school) *Jan.2021*
- National scholarship (award percentage was 0.2%) *Dec.2020*
- Hebei provincial Merit Student (award percentage was 0.14%) *May.2020*
- National Bronze Award in the 5th “Internet+” Innovation and Entrepreneurship Competition (ranking third) *Dec.2019*
- First Prize of Hebei Division of National College Students Mathematical Modeling Competition (ranking first) *Oct.2019*



ACTIVITIES

- Invited by the **Chinese Association of Automation**, I delivered a technical presentation at the World Intelligent Manufacturing Conference. I engaged in in-depth discussions on humanoid robot industrialization with **Xingxing Wang, CEO of Unitree Robotics**, and **chairpersons of publicly listed companies**, and presented research outcomes to the **Governor of Jiangsu Province**. I also presented my work at academic conferences AIM in Seattle, USA.

RELEVANT INFORMATION

Ph.D. Thesis Proposal Title : Research on Semantic Planning and Multi-Layer Collaborative Control Methods for Humanoid Robots.

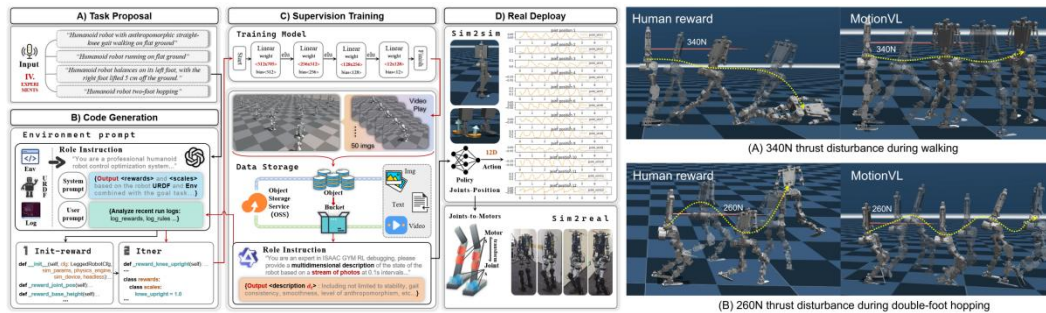
Achieved results1: A new control method DCM-TVDW is proposed for dynamic balance control under time-varying disturbances, and the lower limb balance stability is experimentally verified.

(Top-tier journals in Zone 1: Science China-Technological Sciences)



Achieved results2: Large Language Model(LLM) generative reward function is trained with reinforcement learning(RL) in Isaac Gym , combined with visual language model(VLM) for supervision, and finalized for training and end-side deployment of humanoid straight-knee walking. The fusion of LLM and RL allows robots to learn different skills autonomously, reduces the threshold of reward function design, and improves training efficiency.

(IEEE Robotics and Automation Letters)



Achieved results3: Two teleoperation modes, VR and low-cost master-slave, are implemented, and algorithms such as ACT, $\Pi 0$, and Octo are reproduced and ported, and improved. Collected more operation data, completed the complete process of cleaning and training, and achieved some generalizability. Vision-based 6D position grabbing is also realized, and the inference ability of VLM is utilized to have better operation capability, and VLN (Vision-Language Navigation) .

